INTERIM FEDERAL SPECIFICATION

STYLUS, DUPLICATING STENCIL

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington 25, D.C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward any recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers styli for cutting mimeograph stencils.

1.2 Classification.

1.2.1 Types, styles, and sizes. Styli shall be of the following types, styles, and sizes, as specified (see 6.2):

Type I. Single pointed styli.

Style A. - Needle point.
Style B. - Ball point, fine, non-rotating.
Style C. - Ball point, medium, non-rotating.
Style D. - Ball point, blunt, non-rotating.
Style E. - Curved point, for lettering guides under 3/8 inch.
Style F. - Curved point, for lettering guides 3/8 inch and over.
Style G. - Ruling wheel, for dotted lines.
Size 1. - Heavy line.
Size 2. - Fine line.
Style H. - Shading wheel, for gray tone lines.
Size 1. - Heavy line.
Size 2. - Medium line.
Size 3. - Fine line.
Style I. - Ball point, medium, rotating.
Style J. - Ruling wheel, dot-dash.

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Style K. - Wire loop point.
Size 1. - Heavy line.
Size 2. - Medium line.
Size 3. - Fine line.
Style L. - Screen plate loop.

Type II. Double pointed styli.

Style A.-One end with curved point for lettering guides under 3/8 inch; other end with curved point for lettering guides 3/8 inch and over.
Style B.-One end with ball point, blunt, non-rotating; other end with screen plate loop.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification:

Federal Specifications:

GG-D-818 - Duplicating Machines, Stencil Process.
TT-I-564 - Ink, Stencil, Duplicating Machine, Black, (For Use On Single-Cylinder, Internally Inked, Stencil-Duplicating Machines).
UU-P-388 - Paper, Mimeograph.
UU-P-545 - Paper; Stencil, Duplicating Machine, Dry-Process.
PPP-B-566 - Boxes; Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes; Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material. Material used in the fabrication of styli shall be of a good commercial quality suitable for the purpose.

3.1.1 Steel. Points, wheels, and loops of styli shall be fabricated of steel of a type and hardness which will perform satisfactorily for the purpose intended.

3.1.2 Plastic. Plastic shall be of suitable hardness and rigidity, and shall have low moisture absorption under wide humidity and temperature conditions. The plastic shall retain its design contours without warping, crazing, cracking, or discoloring in service or in storage, and shall withstand the exposure test described in 4.5.1 without change in appearance or in dimensions that will affect serviceability. Plastic used in the stylus handle shall remain straight and rigid after having been subjected to the exposure test described in 4.5.2, and shall be nonflammable, flame resistant, or slow burning when tested as described in 4.5.3.

3.2 Construction.

3.2.1 General. Styli shall be similar in appearance to figure 1. Handles shall be round or square, with round non-slip grips. Points of styli shall be similar in design and construction to the styles shown in figure 1. The point shall be held securely in the handle to form an integral unit. The styli shall have sufficient strength to withstand the test specified in 4.5.4 without damage to the point or handle.

3.2.2 Type I. Type I styli shall have 1 stylus point.

3.2.3 Type II. Type II styli shall have a stylus point on each end of handle. Non-slip grips shall be on both ends of handle. Style A shall have type I, styles E and F stylus points. Style B shall have type I, styles D and L stylus points.

3.3 Performance. Styli shall be tested as specified in 4.5.1. All reproduced copies shall be legible and indicate that the stencil forming material has been cleanly removed where ink transmission through the stencil is required. Thickness and appearance of lines of reproduced copies shall conform generally to illustrations for each style shown on figure 1.

3.4 Dimensions. The overall length of styli shall be not less than 6-1/4 inches. The diameter or width of styli handle at widest point shall be not less than five sixteenths of an inch.

3.5 Temperature conditions. Test samples of styli shall meet all temperature exposure requirements when tested as specified in 4.5.2.

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3.6 Finish. Styli shall be finished in accordance with the manufacturer's standard practice for this type of equipment.

3.7 Marking. Each stylus shall bear the manufacturer's name or trade-mark cast integral with, stamped, or otherwise permenantly marked thereon.

3.8 Workmanship. Styli shall be free from burrs, sharp or rough edges, or other defects which affect appearance or which might affect serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier's responsibility. Unless otherwise specified (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All styli of the same type, style, and size offered for delivery at one time shall be considered a lot for purposes of inspection.

4.3 Sampling for lot acceptance.

4.3.1 Sampling for examination of preparation for delivery. Sampling shall be in accordance with Military Standard MIL-STD-105, inspection level L-5. The sample unit shall be one fully packed shipping container.

4.3.2 Sampling for examination of styli. A random sample of styli shall be selected from each lot offered to the Government in accordance with Military Standard MIL-STD-105, inspection level I. The sample unit shall be one stylus.

4.3.3 Sampling for tests. A random sample of styli shall be selected from each lot in accordance with Military Standard MIL-STD-105, inspection level L-2. The sample unit shall be one stylus.

4.4 Examination.

4.4.1 Examination of preparation for delivery. Samples selected in accordance with 4.3.1 shall be examined to determine compliance with the packaging, packing, and marking requirements of Section 5 of this specification. The acceptable quality level (AQL) shall be 4.0 percent defective.

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4.4.2 Examination of styli. Sample styli, selected in accordance with 4.3.2, shall be examined for defects listed in table I. Any stylus in the sample containing one or more defects shall be rejected. If the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The acceptable quality level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent for minor defects.

Table I - Classification of defects

Defects	Category	
	major	minor
Not type, style, or size specified.	X	
Overall length less than 6-1/4 inches	X	
Diameter or width of handle less than 5/16 inch	X	
Stylus point not secure in handle	X	
Stylus point malformed or broken	X	
Marking not as specified.		X
Workmanship:		
Not free from burrs and sharp or rough edges	X	
Any defects which:		
Affect appearance		X
Impair serviceability	X	

4.4.3 Examination of test results. Sample styli selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Failure to pass one or more of the tests shall render the sample unit defective. The acceptable quality level (AQL) shall be 4.0 percent defective.

4.5 Tests.

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4.5.1 Performance test.

4.5.1.1 Cutting of stencil. A paper stencil conforming to Specification UU-F-545 shall be cut for the various styles as described below by applying only sufficient pressure on the stylus to displace enough stencil forming material from the stencil paper to permit ink transmission during the duplicating process.

Styles B, C, I, and K. A signature with capital letters not less than one half of an inch high and lower case letters not less than one quarter of an inch high shall be impressed on the stencil paper.

Styles A, E, F, G, H, and J. Not less than 5 each 6 inch lines, 60 degree angles, 90 degree angles, and dimension arrows shall be impressed in the stencil paper.

Styles D and L. Using a 20 per-cent halftone Ben Day pattern screen plate for style D stylus and a heavy splatter pattern screen plate for the Style L, the stencil shall be perforated with a 1 inch square by each stylus point.

4.5.1.2 Reproduction from stencil. The cut stencil shall be placed on any type of duplicating machine specified in Specification GG-D-818 and not less than 100 copies shall be reproduced using ink conforming to Specification TT-I-564, type IV, class 1, and paper conforming to Specification UU-P-388, type III, 17 x 22 -- 1000 sheets, 40 pound.

4.5.2 Exposure test. The sample stylus shall be exposed to a minus 80°F. temperature for 72 hours. Remove and examine for compliance with 3.1.2. The stylus shall then be exposed to a temperature of plus 160°F. for 72 hours. Remove and examine for compliance with 3.1.2.

4.5.3 Flammability test. A draft shield shall be used to reduce air current effects. Hold the specimen horisontally within the shield above a Bunsen burner having a 1-inch to 1-1/4-inch blue flame just touching the end of the handle until the specimen ignites, or for a period of not over 30 seconds. Remove the flame. If the specimen burns when in contact with the flame but will not continue to burn when the flame is removed, it is flame resistant. If the specimen continues to burn at a rate of less than 2.5 inches per minute, it is slow-burning. If the specimen bursts into flame immediately, it shall be considered as not meeting the requirements of this specification.

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4.5.4 Strength test. A sheet of paper shall be placed on a platform scale. The sample stylus shall be held between 1-1/4 and 1-1/2 inches from the writing end with point in normal cutting position and at an angle of 50° from the horizontal without the hand touching the paper. A force of 6 pounds shall be applied gradually and uniformly in order to avoid variations and shock within a period of approximately 5 seconds and immediately released. Examine for compliance with 3.2.1

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of levels of preservation, packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A and B. Each stylus shall have the points protected with a sleeve of chipboard tubing and shall be individually packaged in a close-fitting plastic or paperboard tube. One end of the tube shall have a permanent type closure. The open end of the tube shall be provided with a screw cap, friction plug or friction cap closure.

5.1.2 Level C. Each stylus shall be individually packaged in accordance with the supplier's commercial practice.

5.1.3 Intermediate packaging. Ten styli of one type, style and size shall be intermediate packaged in a box conforming to PPP-B-676, type optional, PPP-B-566, style II, type A, class a, pye locks, or to PPP-B-665, style optional.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Styli of one type, style and size shall be packed in boxes conforming to PPP-B-636, class 2, V3c, style optional. The boxes shall be closed, waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B. Ten intermediate packages (100) of one type, style and size stylus shall be packed in boxes conforming to PPP-B-636, type I, class 1. The boxes shall be closed and sealed in accordance with the appendix to the box specification.

5.2.3 Level C. The styli shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military Requirements. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Styli covered by this specification are intended for the following uses:

Style A - Artists use.

B - Signatures, free-hand music, shorthand, maps, tracings.

C - Tracing outlines.

D - Fine screen plate work.

E and F - Lettering guides.

G - Dotted lines; form work.

H - Gray-tone lines; ruled forms and large shaded areas.

I - Signatures, drawing, tracing, etc.

J - Dot-dash lines.

K - General use, signatures, ruling, tracing drawings, etc.

L - Screen plates.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and size desired (see 1.2.1).
(c) Inspection responsibility (for civil agencies only) if other than as specified herein; and administrative provisions for inspection records, when applicable (see 4.1).
(d) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).

6.3 Scope of illustrations. THE ILLUSTRATIONS SHOWN HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of styli which are otherwise in accordance with the requirements of this specification.

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6.4 Federal specifications do not include all types, classes, grades, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

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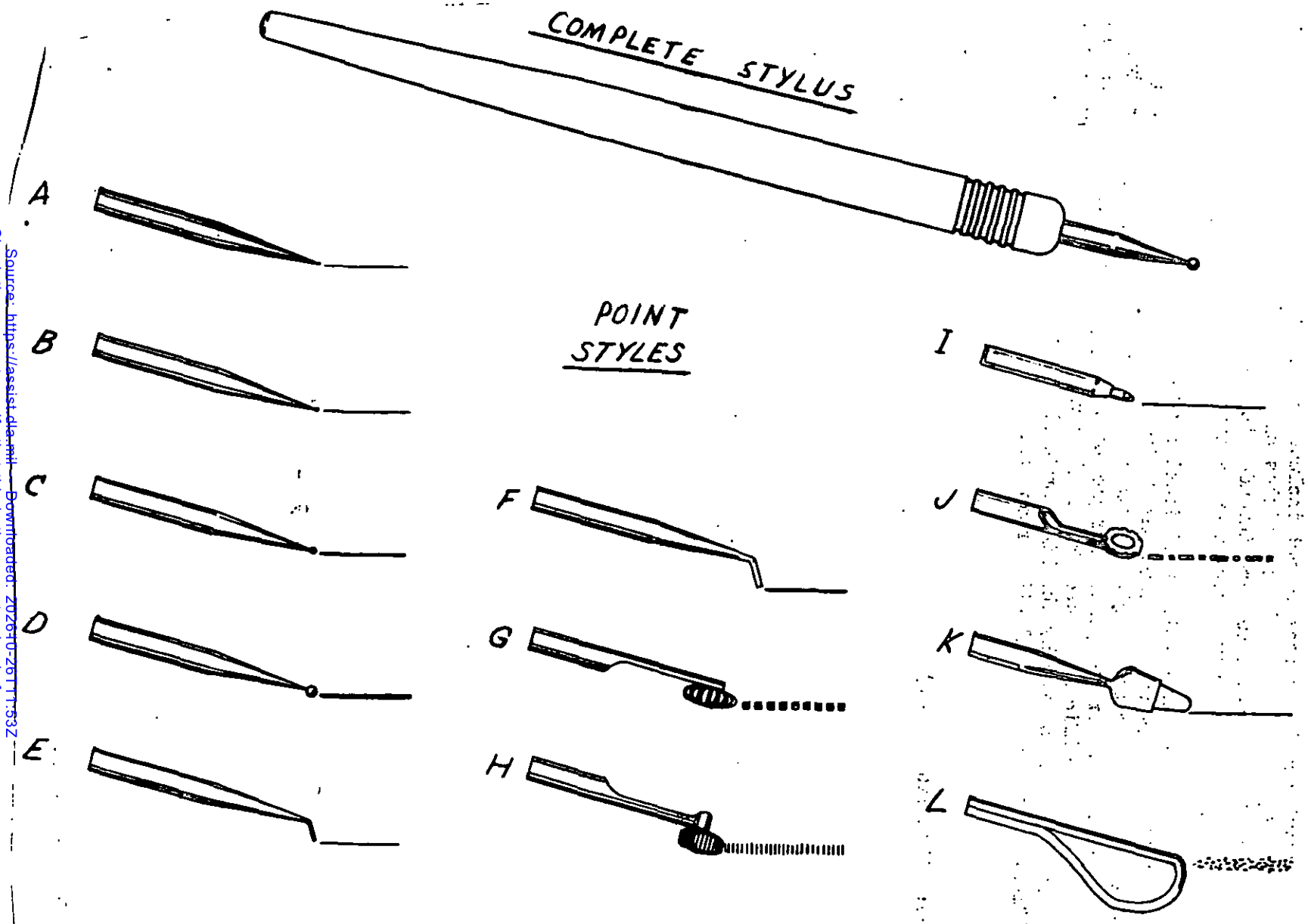


FIGURE-1 STYLI, DUPLICATING STENCIL

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